MudWatt Bioelectric League

Turn a cup of mud into a living battery

At a glance: Can your team make mud generate electricity and defend your design with data? Plan about 80 minutes for the core build-and-baseline block, then short daily check-ins to log readings; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: microbial fuel cells, electrons, decomposition, bioenergy (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- carbon-felt cut into 4×4 in squares: 2 per cell (8 for 4 teams, plus spares)
- clear 24 oz deli containers: 1 per cell (4, plus spares)
- 100 kΩ resistors and alligator-clip leads: 1 resistor and 3 to 4 leads per cell
- multimeters set to DC millivolts: 1 per team (4, plus 2 spare; 2 AAA batteries)
- pond mud or soil samples (collect organic-rich pond or forest mud a day or two ahead)
- nitrile gloves and a trash-bag drip liner per station
- measured sugar cubes (banana peel, stale bread, or leaf litter as alternatives)
- labels and masking tape
- handwashing supplies

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up one station per team with the materials above, sorted and ready.

02 Stage the scoring sheet and any reference values before testing begins.

03 Put out gloves and one trash-bag drip liner per station before any mud is handled. No goggles are needed; this is a damp-mud build with no chemical or splash hazard.

04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Can your team make mud generate electricity and defend your design with data?" Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Baseline reading

Each team wires the 100 kΩ resistor and multimeter, reads the voltage across the resistor in DC millivolts, and logs this day-1 baseline (often only tens of millivolts). Readings climb over the week as the biofilm grows, so treat a rising daily trend, not a fixed number, as the success signal.

Phase 6 Control a variable and log daily

Each team picks ONE variable to hold across the week (fuel type, fuel amount, moisture, or anode packing), changes only that, predicts the effect, and re-reads the voltage each day to compare against the baseline and find the peak.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and on the last day empty the mud into a disposal bag, rinse the felt and containers, bin the liner and wipe the bench, return the meters, leads, and resistors to the bin, then gloves off and wash hands.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: No measurable voltage on day 1. Fix: This is normal; the biofilm needs days to grow. Confirm the two electrodes are not touching, the resistor is in line, the meter is set to DC millivolts, and the anode is pressed flat against the mud with no air trapped under it, then log the low baseline and check again the next day.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to electrogenic bacteria.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which design change moved your readings the most, and how do you know?
- Why must the anode stay buried and the cathode stay in air?
- How is your weekly log like the work of an energy engineer?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Peak voltage reading	35
Daily data log quality	20
Controlled-variable design	15
Redesign or maintenance plan	15
Final explanation	15
Total	**100**

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES soil microbial fuel cell electrode and voltage references (carbon-felt anodes; terrestrial MFC literature on 300 to 600 mV output as the biofilm matures).

Daily voltage log

Each day, set the multimeter to DC millivolts and read the voltage across the 100 kΩ resistor at the same time; the cell is weak on day 1 and climbs as the biofilm grows. Defend the design with the trend, not one number.

Team ____________________ The ONE variable we are testing ____________________

DAY	TIME	VOLTAGE (mV)	NOTES
Mon Jun 22			
Tue Jun 23			
Wed Jun 24			
Thu Jun 25			
Fri Jun 26			
Peak reading			**Day of peak**